

1. Potassium permanganate acts as an oxidant in neutral, alkaline as well as acidic media. The final products obtained from it in the three conditions are, respectively

- (a) $\text{MnO}_2, \text{MnO}_2, \text{Mn}^{2+}$
- (b) $\text{MnO}_4^{2-}, \text{Mn}^{3+}, \text{Mn}^{2+}$
- (c) $\text{MnO}_2, \text{MnO}_4^{2-}, \text{Mn}^{3+}$
- (d) $\text{MnO}, \text{MnO}_4, \text{Mn}^{2+}$

Answer: A — (a) $\text{MnO}_2, \text{MnO}_2, \text{Mn}^{2+}$

Chemical reason: Permanganate is reduced to different manganese species depending on pH. In neutral medium Mn(VII) gives MnO_2 , in strongly alkaline medium it gives manganate, MnO_4^{2-} , and in acidic medium it is reduced to Mn^{2+} . Therefore the option matching those products is correct.

2. In acidic medium one mole of MnO_4^- accepts how many moles of electrons in a redox process ?

- (a) 1
- (b) 3
- (c) 5
- (d) 6

Answer: C — (c) 5

Chemical reason: In acidic medium Mn in MnO_4^- changes from +7 to +2. The decrease of 5 oxidation-number units means one MnO_4^- ion accepts 5 e^- , so one mole of permanganate accepts 5 moles of electrons.

3. In acidic medium potassium dichromate acts as an oxidant according to the equation, $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$. What is the equivalent weight of $\text{K}_2\text{Cr}_2\text{O}_7$? (mol. Wt. = M)

- (a) M
- (b) $M/2$
- (c) $M/3$
- (d) $M/6$

Answer: D — (d) $M/6$

Chemical reason: In acidic medium $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$. Thus one mole of $\text{K}_2\text{Cr}_2\text{O}_7$ accepts 6 moles of electrons, so its n-factor is 6 and its equivalent weight is $M/6$.



4. The correct formula of permanganic acid is

- (a) HMnO_4
- (b) HMnO_5
- (c) H_2MnO_4
- (d) H_2MnO_3

Answer: A — (a) HMnO_4

Chemical reason: Permanganic acid is the oxoacid corresponding to the permanganate ion, MnO_4^- . Adding one proton gives the neutral acid HMnO_4 .

5. Acidified potassium dichromate is treated with hydrogen sulphide. In the reaction, the oxidation number of chromium

- (a) Increases from +3 to +6
- (b) Decreases from +6 to +3
- (c) Remains unchanged
- (d) Decreases from +6 to +2

Answer: B — (b) Decreases from +6 to +3

Chemical reason: In dichromate, chromium is in the +6 state. Acidified dichromate oxidizes H_2S while chromium itself is reduced to Cr^{3+} , so the oxidation number of chromium decreases from +6 to +3.

6. When KMnO_4 reacts with acidified FeSO_4

- (a) Only FeSO_4 is oxidised
- (b) Only KMnO_4 is oxidised
- (c) FeSO_4 is oxidised KMnO_4 and is reduced
- (d) None of these

Answer: C — (c) FeSO_4 is oxidised KMnO_4 and is reduced

Chemical reason: Acidified KMnO_4 is a strong oxidizing agent. Fe^{2+} from FeSO_4 is oxidized to Fe^{3+} , while MnO_4^- is reduced to Mn^{2+} . Hence FeSO_4 is oxidized and permanganate is reduced.



7. When calomel reacts with NH_4OH , we get

- (a) HgNH_2Cl
- (b) $\text{NH}_2\text{-Hg-Hg-Cl}$
- (c) Hg_2O
- (d) HgO

Answer: A — (a) HgNH_2Cl

Chemical reason: Calomel, Hg_2Cl_2 , reacts with NH_4OH to give a black mixture containing finely divided Hg and HgNH_2Cl . The characteristic amidochloride product is therefore HgNH_2Cl .

8. AgCl dissolves in a solution of NH_3 but not in water because

- (a) NH_3 is a better solvent than H_2O
- (b) Ag^+ forms a complex ion with NH_3
- (c) NH_3 is a stronger base than H_2O
- (d) The dipole moment of water is higher than NH_3

Answer: B — (b) Ag^+ forms a complex ion with NH_3

Chemical reason: AgCl is sparingly soluble in water, but NH_3 binds Ag^+ strongly to form the soluble complex $[\text{Ag}(\text{NH}_3)_2]^+$. Complex formation lowers free Ag^+ concentration and drives AgCl dissolution.

9. In solid $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ copper is co-ordinated to

- (a) Five water molecules
- (b) Four water molecules
- (c) One sulphate anion
- (d) One water molecule

Answer: B — (b) Four water molecules

Chemical reason: In crystalline $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, Cu^{2+} is directly coordinated by four water molecules in the equatorial plane; the remaining water molecules are present through axial/ lattice association. Hence four waters are directly coordinated in the usual structural description.



10. A white powder soluble in NH_4OH but insoluble in water is

- (a) BaSO_4
- (b) CuSO_4
- (c) PbSO_4
- (d) AgCl

Answer: D — (d) AgCl

Chemical reason: AgCl is a white, water-insoluble precipitate. It dissolves in aqueous ammonia because Ag^+ forms the soluble diamminesilver(I) complex, $[\text{Ag}(\text{NH}_3)_2]^+$.

11. Verdigris is

- (a) Basic copper acetate
- (b) Basic lead acetate
- (c) Basic lead
- (d) None of these

Answer: A — (a) Basic copper acetate

Chemical reason: Verdigris is the green patina historically identified mainly as basic copper acetate, produced when copper is exposed to acetic-acid vapours and air.

12. Number of moles of $\text{K}_2\text{Cr}_2\text{O}_7$ reduced by one mole of Sn^{2+} ions is

- (a) $1/3$
- (b) 3
- (c) 1.6
- (d) 6

Answer: A — (a) $1/3$

Chemical reason: Sn^{2+} is oxidized to Sn^{4+} and therefore loses $2 e^-$ per mole. One mole of $\text{Cr}_2\text{O}_7^{2-}$ requires $6 e^-$ for reduction to Cr^{3+} , so 3 mol Sn^{2+} reduce 1 mol dichromate; therefore 1 mol Sn^{2+} reduces $1/3$ mol $\text{K}_2\text{Cr}_2\text{O}_7$.



13. Which one of the following is reduced by hydrogen peroxide in acid medium

- (a) Potassium permanganate
- (b) Potassium iodide
- (c) Ferrous sulphate
- (d) Potassium ferrocyanide

Answer: A — (a) Potassium permanganate

Chemical reason: In acidic solution H_2O_2 can act as a reducing agent and is oxidized to O_2 . Acidified KMnO_4 accepts electrons readily, so permanganate is the species that is reduced.

14. Which oxide of manganese is amphoteric

- (a) MnO_2
- (b) Mn_2O_3
- (c) Mn_2O_7
- (d) MnO

Answer: A — (a) MnO_2

Chemical reason: MnO_2 is amphoteric: it can react with strong acids and, under strongly basic/oxidizing conditions, can form manganate⁻ type species. This intermediate oxidation state gives it both acidic and basic behaviour.

15. Which one of the following oxides is ionic

- (a) MnO
- (b) Mn_2O_7
- (c) CrO_3
- (d) P_2O_5

Answer: A — (a) MnO

Chemical reason: MnO contains Mn^{2+} and O^{2-} and has predominantly ionic bonding, unlike higher manganese oxides which show increasing covalent character as the oxidation state rises.



16. Correct formula of calomel is

- (a) Hg_2Cl_2
- (b) HgCl_2
- (c) $\text{HgCl}_2 \cdot \text{H}_2\text{O}$
- (d) HgSO_4

Answer: A — (a) Hg_2Cl_2

Chemical reason: Calomel is mercurous chloride. Mercury(I) exists as the dimeric ion Hg_2^{2+} , so the chloride formula is Hg_2Cl_2 .

17. One of the important use of ferrous sulphate is in the

- (a) Manufacture of blue black ink
- (b) Manufacture of chalks
- (c) Preparation of hydrogen sulphide
- (d) Preparation of anhydrous ferric chloride

Answer: A — (a) Manufacture of blue black ink

Chemical reason: Ferrous sulfate is used in making traditional blue-black inks because $\text{Fe}^{2+}/\text{Fe}^{3+}$ reacts with tannins (such as gallic/tannic acid derivatives) to form dark iron complexes.

18. Copper sulphate is not used

- (a) In electrotyping
- (b) In dyeing and calicoprinting
- (c) In detecting water
- (d) As fertilizer

Answer: D — (d) As fertilizer

Chemical reason: The application follows from the characteristic chemical property of the material—such as catalytic activity, stability, conductivity, or corrosion resistance. That property is specifically associated with As fertilizer.



**19. Blue vitriol is**

- (a) CuSO_4
- (b) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
- (c) Cu_2SO_4
- (d) $\text{CuSO}_4 \cdot \text{H}_2\text{O}$

Answer: B — (b) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

Chemical reason: Blue vitriol is the common name for hydrated copper(II) sulfate, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$. Its blue colour comes from hydrated Cu^{2+} ions.

20. A solution of copper sulphate may be kept safely in the container made up of

- (a) Fe
- (b) Ag
- (c) Zn
- (d) Al

Answer: B — (b) Ag

Chemical reason: The correct choice is Ag. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

